NOT Gate

Design File (not_gate.vhd)

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity not_gate is
    Port (
        A : in  STD_LOGIC;
        Y : out STD_LOGIC
    );
end not_gate;

architecture simple of not_gate is
begin
    Y <= NOT A;
end simple;
```

Testbench File (not_gate_tb.vhd)

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity not_gate_tb is
end not_gate_tb;

architecture test of not_gate_tb is

    component not_gate
        Port (
            A : in  STD_LOGIC;
            Y : out STD_LOGIC
        );
    end component;

    signal A : STD_LOGIC;
    signal Y : STD_LOGIC;

begin

    uut: not_gate
        port map (
            A => A,
            Y => Y
        );

    process
    begin
        A <= '0';
        wait for 10 ns;

        A <= '1';
        wait for 10 ns;

        wait;
    end process;

end test;
```

AND Gate

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity basic_gate is
    Port (
        A : in  STD_LOGIC;
        B : in  STD_LOGIC;
        Y : out STD_LOGIC
    );
end basic_gate;

architecture simple of
basic_gate is
begin
    Y <= A AND B;
end simple;
```

NAND Gate

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity basic_gate is
    Port (
        A : in  STD_LOGIC;
        B : in  STD_LOGIC;
        Y : out STD_LOGIC
    );
end basic_gate;

architecture simple of
basic_gate is
begin
    Y <= A NAND B;
end simple;
```

OR Gate

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity basic_gate is
    Port (
        A : in  STD_LOGIC;
        B : in  STD_LOGIC;
        Y : out STD_LOGIC
    );
end basic_gate;

architecture simple of
basic_gate is
begin
    Y <= A OR B;
end simple;
```

NOR Gate

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity basic_gate is
    Port (
        A : in  STD_LOGIC;
        B : in  STD_LOGIC;
        Y : out STD_LOGIC
    );
end basic_gate;

architecture simple of
basic_gate is
begin
    Y <= A NOR B;
end simple;
```

XOR Gate

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity basic_gate is
    Port (
        A : in  STD_LOGIC;
        B : in  STD_LOGIC;
        Y : out STD_LOGIC
    );
end basic_gate;

architecture simple of
basic_gate is
begin
    Y <= A XOR B;
end simple;
```

XNOR Gate

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity basic_gate is
    Port (
        A : in  STD_LOGIC;
        B : in  STD_LOGIC;
        Y : out STD_LOGIC
    );
end basic_gate;

architecture simple of
basic_gate is
begin
    Y <= A XNOR B;
end simple;
```

Test Bench Code :

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity gate_tb is
end gate_tb;

architecture test of gate_tb is

    component basic_gate
        Port (
            A : in  STD_LOGIC;
            B : in  STD_LOGIC;
            Y : out STD_LOGIC
        );
    end component;

    signal A : STD_LOGIC;
    signal B : STD_LOGIC;
    signal Y : STD_LOGIC;

begin

    uut: basic_gate
        port map (
            A => A,
            B => B,
            Y => Y
        );

    process
    begin
        A <= '0'; B <= '0';
        wait for 10 ns;

        A <= '0'; B <= '1';
        wait for 10 ns;

        A <= '1'; B <= '0';
        wait for 10 ns;

        A <= '1'; B <= '1';
        wait for 10 ns;

        wait;
    end process;

end test;
```